FLAIR: A Federated ICU Benchmark for External Validation on the Common Longitudinal ICU Format

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Abstract

Background. Clinical decision-support models for the intensive care unit (ICU) are produced far faster than the multi-site evidence needed to judge whether they work in hospitals that did not build them. Public ICU benchmarks accelerate development but centralize their test data, so they inherit one source's biases and can saturate under repeated tuning. Traditional external validation requires bilateral data-use agreements and patient-level data transfer, and rarely scales beyond a few sites.

Methods. FLAIR is an open-source federated benchmark and model-to-data protocol for ICU prediction. It is built on the Common Longitudinal ICU Format (CLIF), a critical-care data standard with shared tables and mCIDE categories. Developers build against public MIMIC-CLIF; after Benchmark Committee review, byte-identical code runs locally at participating CLIF sites, and only aggregate, cell-suppressed report bundles return. Five binary tasks span mortality, long-term acute-care discharge, extubation failure, CDC Adult Sepsis Event, and ICU readmission. A single frozen XGBoost count-feature model serves as the reference baseline, evaluated on a held-out

MIMIC-CLIF split and at three external health systems (278,065 ICU stays in total).

Results. Trained on MIMIC-CLIF, the baseline reached strong internal discrimination (AUROC 0.64–0.84; landmark tasks scored by pooled dynamic AUC) but degraded at every external site, and the loss was strongly task-dependent: the median external AUROC fell only 0.067 for daily mortality yet 0.257 for sepsis, which dropped below chance at all three external sites (0.430–0.449). Extubation failure was heterogeneous—matching the internal value at one site (0.703) but falling to 0.531 at another—a spread a single pooled score would conceal.

Conclusions. FLAIR makes multi-site ICU evaluation reproducible, auditable, and feasible without routine patient-level data transfer. It is a benchmark and protocol resource, not evidence of clinical utility for any specific model.

1 Introduction

The bottleneck in ICU clinical AI is no longer the supply of models; it is trustworthy evidence that a model works somewhere it was not built. Predictive algorithms for sepsis, extubation readiness, mortality, and hospital operations appear continually, but whether a given model behaves acceptably outside its development hospital is often unknown. That evidence gap is, in large part, an access gap: representative multi-institutional ICU data are rarely available to the team that needs to test a model.

External validation of flagship electronic health record (EHR)-based decision support has stayed narrow and slow. The Epic Sepsis Model, deployed across many United States hospitals, was first independently evaluated at a single site, where discrimination fell to an AUC of 0.63 against vendor-claimed ranges of 0.76 to 0.83 (Wong et al., 2021; Habib et al., 2021). Duke's Sepsis Watch and the Targeted Real-time Early Warning System each required years and reached only a handful of sites (Sendak et al., 2025; Adams et al., 2022); Mayo Clinic's electrocardiogram screen needed multi-year coordination to cover several external sites (Carter et al., 2025). Serious validation is possible, but it does not keep pace with model

production.

Public benchmarks help but cannot close the gap alone. ICU benchmarks built on MIMIC, eICU, HiRID, and AUMCdb are essential for reproducible development (van de Water et al., 2023; Yèche et al., 2021; Pollard et al., 2018), yet they centralize their test data, inherit its selection biases (Röösli et al., 2022), and can make performance look more transportable than it is once the same public set is tuned against repeatedly (Recht et al., 2019; Chekroud et al., 2024). This is precisely the "why not just use an existing benchmark" question: EHRSHOT and the FEMR tokenization tradition offer a public few-shot benchmark that FLAIR builds on for its baseline representation (Wornow et al., 2023; Steinberg et al., 2021), and the Medical Event Data Standard offers a data standard (Arnrich et al., 2025), but none provides a standing, ICU-scoped, *federated validation* layer over data that never leaves the hospital.

Several communities show pieces of a better approach. OHDSI and LEGEND federate studies across OMOP (Reps et al., 2020); MedPerf orchestrates model-to-data imaging benchmarks (Karargyris et al., 2023); the FeTS Challenge federated imaging *training* (Pati et al., 2022); OpenSAFELY moves analytic code to national primary-care data without moving rows (Collaborative, 2021); and DREAM and EHR DREAM turned model-to-data evaluation into a challenge format (Guinney and Saez-Rodriguez, 2018; Bergquist et al., 2024). What has been missing for ICU EHR decision support is the combination of a critical-care data standard, a reusable task catalog, and a standing federated protocol.

FLAIR (Federated Learning Assessment for ICU Research) supplies that combination. The idea is simple: the model goes to the data, and aggregate evaluation reports come back. A developer writes code against MIMIC-CLIF, the public CLIF expression of MIMIC-IV (Johnson et al., 2023, 2016); after review, the same code runs inside each site, which returns tables and metrics, not patient rows. We describe FLAIR as three coupled artifacts—a data layer (CLIF, with MIMIC-CLIF as the public entry point), a task layer (five inspectable Python tasks with an open catalog), and a protocol layer (committee-reviewed, aggregate-only model-to-data execution)—and we show that even a simple reference baseline already exposes

large, task-dependent cross-site degradation. FLAIR does not certify models for bedside use; its purpose is to make multi-site ICU evaluation reproducible, auditable, and feasible without routine patient-level data transfer.

2 Methods

Data layer: CLIF and MIMIC-CLIF

CLIF harmonizes ICU EHR content into a limited set of relational tables with mCIDE category definitions (Rojas et al., 2025), covering patients, hospitalizations, admission-discharge-transfer (ADT) events, vitals, labs, medications, respiratory support, patient assessments, microbiology, and diagnoses. Because task code refers to shared table names and category values, it reduces the site-specific feature engineering otherwise required before a model can be evaluated. In spirit CLIF is to critical care what mCODE is to oncology—a clinically scoped common-data-element layer that makes analytic code portable (HL7 International, 2024)—and it is complementary to broader EHR event standards such as the Medical Event Data Standard (Arnrich et al., 2025). MIMIC-CLIF, the public development dataset, expresses MIMIC-IV in the CLIF v2.1 schema and contains approximately 72,800 ICU stays (Johnson et al., 2023). Developers use it to debug task definitions, train baselines, and confirm that submissions follow the cohort and prediction-file contracts before any federated execution, yielding a develop-on-public, validate-on-private workflow.

Task layer: five ICU tasks

Each task is a single Python module with a `META` dictionary (task class, temporal pattern, prediction unit, label, clinical threshold, direction, report mode) and a `build()` function returning one row per prediction opportunity. Every row carries a stable `prediction_id`, a stitched-hospitalization key, a prediction timestamp (`prediction_dttm`, the no-future-data

boundary), the label, the train/test split, and demographics for reporting. The five tasks (Table 1) span outcome, disposition, respiratory, infectious-deterioration, and utilization questions (Thille et al., 2011; Rojas et al., 2018). Task 4 predicts onset of a CDC Adult Sepsis Event (Rhee et al., 2019) within 6 hours on an hourly hospital-wide grid, using the CLIF Adult Sepsis Event utility with lactate and the `microbiology_culture` table, and stopping strictly before onset; it follows the before-treatment sepsis formulation of Kamran et al. (2024) and contrasts with antibiotic-anchored Sepsis-3 definitions (Singer et al., 2016; Seymour et al., 2016). The catalog is designed to grow: any contributor can propose a task as a module, and acceptance requires MIMIC-CLIF reproducibility, clinician-led face-validity review, and site-governance sign-off at a minimum of two sites—a structural counter to benchmark saturation.

Three methodological safeguards make cohorts comparable across sites. First, fragmented hospitalizations (e.g., emergency-department to inpatient transfer, or discharge and rapid readmission) are stitched into a single unit, `hospitalization_join_id`, using a 6-hour gap rule, so one clinical episode is never split across the train/test boundary. Second, features may use only data at or before `prediction_dttm`. Third, splits are assigned at the stitched-hospitalization grain (75/25): the default is a *temporal* split (oldest 75% train, newest 25% test), which estimates pseudo-prospective performance—how the frozen model would behave going forward in time; MIMIC-CLIF is the exception, using a deterministic hash split because its admission dates are de-identified and carry no usable temporal order. A SHA-256 audit hash over the task source and its policy metadata is stamped into every report, so sites can prove they ran identical task code.

Reference baseline

The reference baseline is one frozen XGBoost model over point-in-time count features, in the EHRSHOT/FEMR tradition (Wornow et al., 2023; Steinberg et al., 2021). CLIF events

Task	Outcome	Prediction grid	Report mode	Threshold	Prev.
T1	ICU daily mortality	daily 07:00	landmark	0.95 (above)	0.125
T2	LTACH discharge	daily 07:00	landmark	0.50 (above)	0.062
T3	Extubation failure 24 h	at extubation	episodic	0.15 (below)	0.054
T4	Sepsis (CDC ASE) 6 h	hourly	peak	0.10 (above)	0.061
T5	Unplanned ICU readmission	at ICU discharge	episodic	0.20 (above)	0.093

Table 1: The five FLAIR benchmark tasks. Prevalence is the MIMIC-CLIF positive rate on the scored test set (first landmark for T1–T2; per stay for T3–T5). Clinical thresholds and directions are single-sourced in the package policy registry and folded into each task's audit metadata. Report modes are defined in the text. Source: FLAIR task code and baseline report artifacts.

are converted to a fixed 211-code Event List Format vocabulary; each prediction row is represented by counts of codes observed strictly before `prediction_dttm`, with numeric and categorical codes truncated so that units are never features and discharge diagnoses excluded to prevent leakage. Hyperparameters are tuned once on MIMIC-CLIF (Optuna, 30 trials, 5-fold cross-validated AUROC; see Appendix). The trained model and token dictionary are then shipped unchanged to each site for inference only; there is no per-site training. These values are benchmark checks, not a claim that the baseline is clinically adequate.

Evaluation modes

Because continuous ICU tasks can be scored in more than one clinically meaningful way, FLAIR separates task construction from scoring mode. *Episodic* scoring is used when there is one prediction per stay (T3, T5). *Landmark* scoring is used for daily trajectories (T1, T2): a prediction is made at each daily landmark among patients still at risk. For a single headline number per task we report the *pooled dynamic AUC*—the global concordance that summarizes the incident/dynamic time-dependent AUC by comparing cases against controls within each landmark risk set and pooling those comparisons across landmarks (Heagerty and Zheng, 2005; van Houwelingen, 2007); the per-landmark AUROC curve is retained in the Appendix. *Peak* scoring is used for continuous screening (T4): it collapses each stay to its highest predicted risk before computing discrimination and subgroup metrics. Calibration

and decision-curve analysis are omitted under peak scoring because taking the maximum of many probabilities distorts calibration. This distinction is load-bearing—a row-level metric can overstate the performance a unit would experience acting on one alarm per stay. All metrics are scored on the held-out test split with percentile-bootstrap 95% confidence intervals resampled at the stay level.

Protocol layer: federated execution and governance

The protocol makes FLAIR's privacy properties operational (Figure 1). A **Model Owner** submits a frozen model, code folder, or container developed against MIMIC-CLIF. The **Benchmark Committee**—clinicians, methodologists, and site data stewards—reviews submissions for reproducibility, clinical face validity, and security, and stamps an audit hash before dispatch. The **FLAIR Portal** registers tasks, models, and versions, distributes byte-identical artifacts, collects reports, and renders the leaderboard. At each institution, a **Site Lead or Data Steward** runs the approved artifact locally against site CLIF tables and returns only the standardized non-PHI report bundle. Three properties define the access model: no patient-level data leave any site; the library enforces suppression (cohort counts below 10 suppressed, subgroups with fewer than 10 observations omitted, sparse calibration bins withheld); and governance review precedes execution, with sites retaining final approval over what runs locally. Each report bundle carries cohort summaries, discrimination, calibration and decision curves where applicable, subgroup and lead-time summaries, a suppressed code registry, and audit metadata (task name, task hash, package version, report mode, split, number of stays, prevalence, and operating threshold). This model-to-data pattern follows the DREAM and EHR DREAM lineage (Guinney and Saez-Rodriguez, 2018; Bergquist et al., 2024); FLAIR's contribution is to make it a standing ICU benchmark. Reporting follows the structure expected of clinical prediction models (Collins et al., 2024), with a Datasheet framework for the public cohort (Gebru et al., 2021) and a Model Card framework for the baseline (Mitchell et al., 2019).

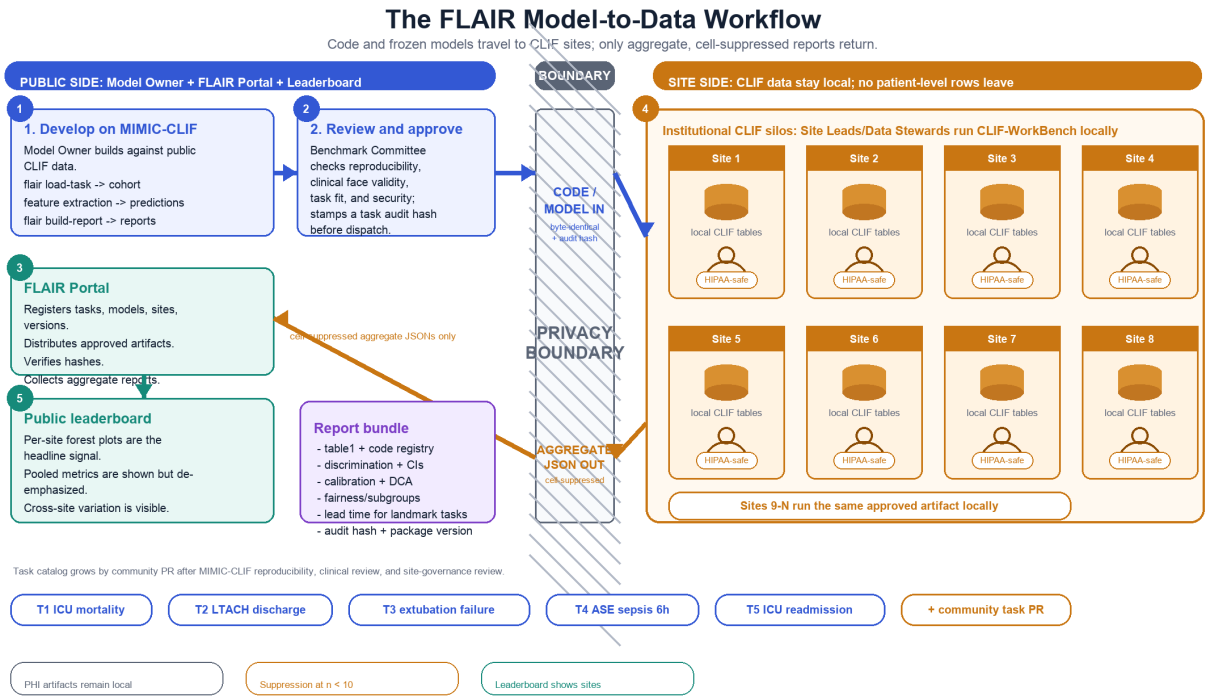


Figure 1: The FLAIR model-to-data loop. A Model Owner develops against public MIMIC-CLIF and submits a frozen model, code, or container. The Benchmark Committee reviews it for reproducibility, clinical face validity, and security, and stamps a task audit hash. The FLAIR Portal distributes byte-identical code to participating CLIF sites, where a Site Lead or Data Steward runs it locally against site CLIF tables. Patient-level cohorts, features, and predictions remain inside the institutional boundary; only aggregate, cell-suppressed JSON report bundles return. The leaderboard foregrounds per-site variation rather than a single pooled scalar. Source: authors.

3 Results

Consortium cohort

Table 2 describes the cohort. Four sites have reported observed federated runs—MIMIC-CLIF (the public development set and internal holdout) and three external academic health systems: Rush, Northwestern Medicine, and the University of Chicago Medicine. Four additional CLIF sites are onboarding and are shown as placeholder rows. The four reporting sites represent 278,065 ICU stays across distinct urban academic and multi-hospital case mixes, with median age from 61 to 67 years and female proportion from 43% to 48%. This heterogeneity is the

Site	ICU stays	Female %	Age	SOFA
MIMIC-CLIF	72,778	43.2	67	5
Rush	53,356	47.8	63	4
Northwestern Medicine	114,335	44.0	66	5
University of Chicago Medicine	37,596	43.5	61	6
Consortium site 5	—	—	—	—
Consortium site 6	—	—	—	—
Consortium site 7	—	—	—	—
Consortium site 8	—	—	—	—

Table 2: Federated consortium cohort characteristics. Age is the median in years; SOFA is the median hospital-maximum score. Four sites have reported observed federated runs; four additional consortium sites are onboarding (shown as placeholder rows) and have not yet returned results. Per-site cell suppression at $n < 10$ is applied. Source: FLAIR baseline report artifacts, single-sourced through `flair_paper_data.py`.

setting in which transportability, rather than internal accuracy, is the relevant question.

Cross-site degradation

The central finding is observed and unambiguous: discrimination that is strong on the MIMIC-CLIF internal holdout does not survive transport to hospitals that did not build the model, and how far it falls depends sharply on the task (Figure 2; Figure 3). The median internal-to-external AUROC drop across the three external sites was 0.067 for daily mortality (internal 0.835; external median 0.768), 0.176 for LTACH discharge (0.761; 0.585), 0.033 for extubation failure (0.703; 0.670), 0.147 for readmission (0.637; 0.490), and 0.257 for sepsis (0.697; 0.440).

Sepsis was the most fragile task: discrimination fell below chance at *all three* external sites (0.430 at Rush, 0.449 at Northwestern, 0.440 at the University of Chicago Medicine), so the reproducible external result is a model that is worse than a coin flip on the task with the greatest clinical stakes. Because Task 4 is a rare, hospital-wide, hourly screen, it is both the most consequential and the most sensitive to distribution shift. Extubation failure showed the opposite lesson: it degraded little at the median but was highly heterogeneous, matching the internal value at Northwestern (0.703) while collapsing to 0.531 at the University of Chicago

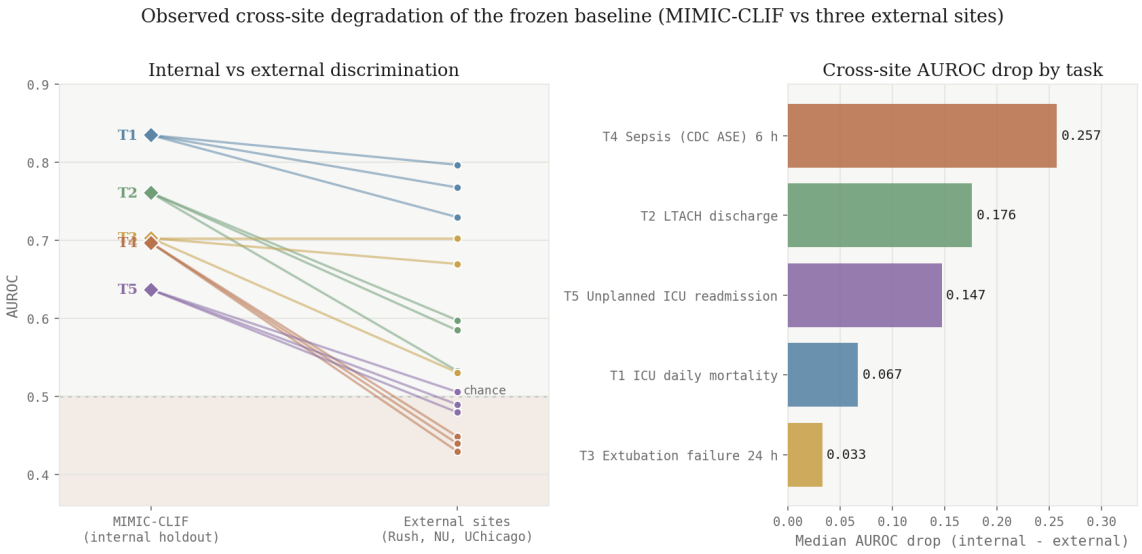


Figure 2: Observed cross-site degradation of the frozen baseline from the MIMIC-CLIF internal holdout to the three external sites (Rush, Northwestern Medicine, University of Chicago Medicine). Left: for each task a line runs from the internal AUROC to each external site; sepsis and readmission cross below the chance band. Right: the median internal-to-external AUROC drop by task, ordered by magnitude. All four sites are observed. Landmark tasks use the pooled dynamic AUC. Source: FLAIR baseline report artifacts.

Medicine. That within-task spread is exactly what a single pooled AUROC across sites would hide and what the per-site forest plot (Figure 3) is designed to surface. Full per-site metrics, calibration, decision curves, lead-time, and subgroup results are provided in the Appendix.

4 Discussion

FLAIR combines three properties that have appeared separately but not together at the scale ICU clinical AI requires: evaluation across heterogeneous hospitals without patient-data movement, a clinically scoped ICU data standard that makes code portable, and a governance protocol that lets the task catalog grow through community contribution. The empirical payoff is already visible in a deliberately simple baseline: strong internal discrimination does not survive the trip to a hospital that did not build the model, and the degradation is task-dependent, with sepsis screening the most fragile. This is consistent with a large

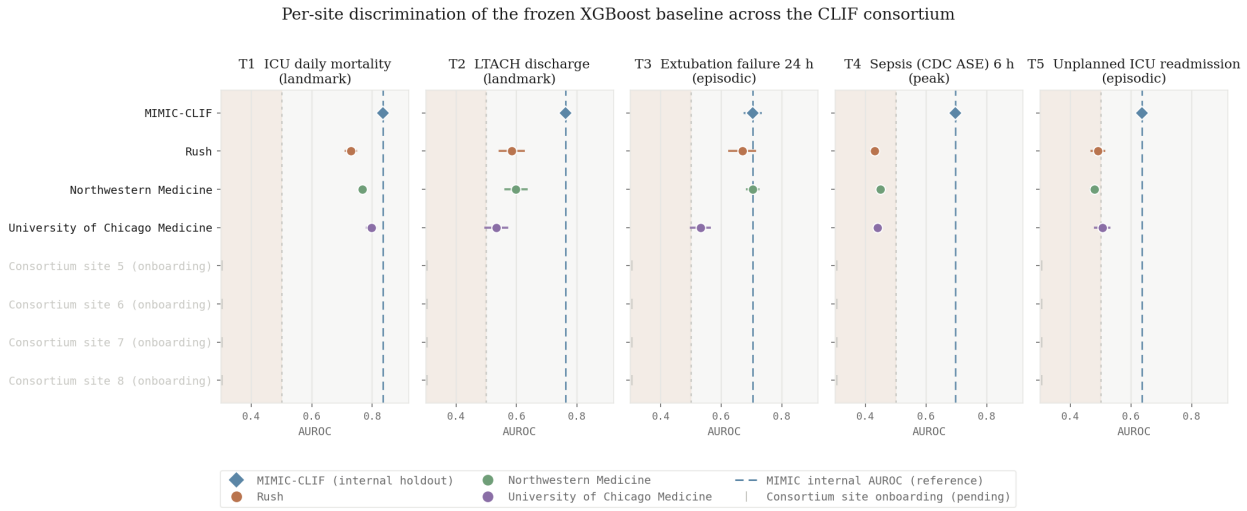


Figure 3: Per-site discrimination of the frozen baseline across the CLIF consortium, one panel per task. Each marker is a site's AUROC with a 95% confidence interval; the dashed line is the MIMIC-CLIF internal reference and the shaded band marks worse-than-chance discrimination. The four observed sites (MIMIC-CLIF, Rush, Northwestern Medicine, University of Chicago Medicine) are shown in colour; four onboarding consortium sites appear as greyed placeholder rows pending federated runs. Landmark tasks (T1, T2) use the pooled dynamic AUC. Source: FLAIR baseline report artifacts, single-sourced through `flair_paper_data.py`.

literature on dataset shift in clinical prediction (Finlayson et al., 2021), but FLAIR turns that observation from a warning into a measurement.

Saturation resistance and evidence posture. Because FLAIR's test data remain siloed at participating sites, the data underlying the leaderboard never enter public hands, which limits the risk that future models are pretrained on the evaluation set and reduces the incentive to tune to a private test set that cannot be seen (Recht et al., 2019; Chekroud et al., 2024). FLAIR also federates *evaluation* rather than *training*: aggregate, cell-suppressed metrics leak less patient information than the model updates exchanged in federated learning, and there is no per-site trained model object to share. By making cohort definitions, exclusions, and labels re-inspectable across sites through byte-identical code and audit hashes, FLAIR provides an evidence layer between local development and any clinician-facing claim. It is not a substitute for prospective trials; for sites that opt in, the same protocol can support prospective silent-trial evaluation (Goodman et al., 2025; Joshi et al., 2025; NEJM AI Pragmatic Trial Authors,

2024).

Limitations. FLAIR depends on CLIF adoption; sites that have not mapped their data need an extract-transform-load step first. The current suite is limited to structured EHR data and binary classification, and does not include notes, waveforms, images, treatment-effect estimation, or prospective bedside impact. Aggregate-only reporting protects privacy but limits fine-grained error analysis, and suppressed small subgroups are not visible centrally. The protocol evaluates frozen models and does not perform federated training. The present results come from four sites; four additional consortium sites are onboarding, and a larger network will sharpen the per-site distributions reported here. Because each external site uses a temporal train/test split, some degradation may reflect temporal as well as cross-site shift.

Conclusion. FLAIR is a federated benchmark and protocol for ICU prediction research, combining a shared critical-care data standard, public development on MIMIC-CLIF, five inspectable tasks, committee-reviewed model-to-data execution, and aggregate-only reporting. Its contribution is not a new model but a reproducible way to ask whether ICU prediction models behave similarly across institutions—and its first, observed answer is that a strong internal model can fail badly at a hospital that did not build it.

Required Statements

Data and Code Availability. The FLAIR source code, task definitions, feature extractors, report-generation library, and manuscript build scripts are released under Apache 2.0 at `https://github.com/clif-consortium/FLAIR`. MIMIC-CLIF is distributed through PhysioNet under the applicable MIMIC data use agreement. Consortium site data are not redistributable and are accessible only through approved local execution. Baseline model archive: [AUTHOR: Zenodo DOI assigned at acceptance]. Container image: `ghcr.io/clif-consortium/flair:v1.0`

[AUTHOR: tag pending release].

Institutional Review. Each participating site executes FLAIR under its local institutional review board approval or exemption for retrospective CLIF participation; central protocol coordination is provided by the University of Chicago [AUTHOR: final IRB numbers pending].

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Author Contributions. [AUTHOR: CRediT roles to be completed at submission].

Conflicts of Interest. [AUTHOR: disclosures to be completed at submission].

AI-Use Disclosure. Large language model assistance was used for manuscript drafting and for the figure- and table-generation code; all content, numbers, and code were reviewed and verified by the authors. [AUTHOR: confirm journal-required wording].

A Supplementary Appendix

S1. Reference baseline details

The baseline converts CLIF event rows into token counts over a fixed 211-code Event List Format vocabulary. For each prediction row, only events strictly before the prediction time are counted, yielding a sparse matrix with one row per prediction and one column per token; missing tokens are zero and tokens unseen in training are ignored at inference. Numeric codes are truncated to two levels and categorical codes to three, so measurement units never become features; discharge diagnoses are excluded to prevent leakage. Age at admission is appended as a single dense feature. Hyperparameters are selected once on MIMIC-CLIF with Optuna (tree-structured Parzen estimator, 30 trials, maximizing 5-fold cross-validated AUROC) over tree depth, learning rate, subsampling, and regularization; class imbalance is handled with `scale_pos_weight`. The final model and vocabulary are frozen and shipped to sites.

S2. Full internal baseline metrics (MIMIC-CLIF)

Task	Mode	n	Prev.	AUROC (95% CI)	AUPRC	Cal. slope	Cal. int.	Brier
T1, ICU daily mortality	landmark	18,195	0.125	0.835 (0.825–0.846)	0.547	1.08	-1.18	0.131
T2, LTACH discharge	landmark	18,195	0.062	0.761 (0.744–0.778)	0.299	0.87	-1.69	0.129
T3, extubation failure 24 h	episodic	6,693	0.054	0.703 (0.678–0.730)	0.131	0.44	-1.82	0.066
T4, sepsis 6 h	peak	19,943	0.061	0.697 (0.680–0.712)	0.132	—	—	—
T5, unplanned ICU readmission	episodic	16,806	0.093	0.637 (0.622–0.652)	0.234	0.64	-1.88	0.150

Table S1: Full observed MIMIC-CLIF internal baseline metrics for the XGBoost count-feature model. Landmark tasks (T1, T2) report the pooled dynamic AUC and pooled calibration across landmarks. Calibration is omitted for T4 by design under peak scoring. Source: FLAIR baseline report artifacts.

S3. Full per-site leaderboard

Site	T1 (95% CI)	T2 (95% CI)	T3 (95% CI)	T4 (95% CI)	T5 (95% CI)
MIMIC-CLIF	0.835 (0.825–0.846)	0.761 (0.744–0.778)	0.703 (0.678–0.730)	0.697 (0.680–0.712)	0.637 (0.622–0.652)
Rush	0.730 (0.713–0.746)	0.585 (0.545–0.623)	0.670 (0.627–0.712)	0.430 (0.416–0.444)	0.490 (0.470–0.511)
Northwestern Medicine	0.768 (0.759–0.779)	0.598 (0.564–0.633)	0.703 (0.685–0.722)	0.449 (0.441–0.458)	0.480 (0.467–0.494)
Univ. of Chicago Medicine	0.797 (0.782–0.812)	0.533 (0.498–0.570)	0.531 (0.499–0.562)	0.440 (0.430–0.452)	0.506 (0.481–0.529)
Consortium site 5	—	—	—	—	—
Consortium site 6	—	—	—	—	—
Consortium site 7	—	—	—	—	—
Consortium site 8	—	—	—	—	—
External median	0.768	0.585	0.670	0.440	0.490

Table S2: Per-site AUROC (95% CI) for the frozen baseline, mode-matched per task; landmark tasks (T1, T2) report the pooled dynamic AUC. Four sites are observed; four consortium sites are onboarding (placeholder rows). External median is over the three observed external sites. Source: FLAIR baseline report artifacts, single-sourced through `flair_paper_data.py`.

S4. MIMIC-CLIF cohort snapshot per task

S5. Comparison with adjacent federated infrastructures

S6. Supplementary figures

Per-task supporting figures generated from the observed site reports are retained as supplementary material: ROC and precision-recall overlays across the four observed sites for

Measure	T1 mortality	T2 LTACH	T3 extubation	T4 sepsis	T5 readmission
Eligible hospitalizations	72,778	72,778	26,770	79,772	67,221
Sex, F/M	31,409/41,369	31,409/41,369	10,032/16,738	35,415/44,357	29,195/38,026
Age, median (IQR)	67 (55–77)	67 (55–77)	66 (56–75)	67 (55–77)	66 (55–77)
SOFA hospital maximum, median (IQR)	5 (2–8)	5 (2–8)	7 (5–10)	4 (2–7)	4 (2–7)
Charlson score, median (IQR)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)

Table S3: Observed MIMIC-CLIF per-task cohort snapshot. Counts are eligible stitched hospitalizations (`hospitalization_join_id`) per task; T1 and T2 share the daily-landmark cohort, while T3–T5 apply task-specific eligibility. Charlson scores are missing for roughly half of stays in the MIMIC-CLIF export. Source: FLAIR baseline report artifacts (per-task `table1.json`).

Feature	OHDSI/ LEGEND	MedPerf	FeTS	Open SAFELY	FLAIR
Data standard	OMOP	task specific	imaging	TPP/EMIS	CLIF
Federation unit	SQL/coefs	container	gradients	analytic code	model/code execution
Main use	network studies	imaging benchmarks	federated training	secure analysis	ICU EHR benchmarks
Hosted-model portal	no	yes	yes	no	yes
Patient rows shared	no	no	no	no	no
Extensible tasks	limited	partial	no	not applicable	yes

Table S4: FLAIR compared with adjacent federated or model-to-data infrastructures (Reps et al., 2020; Karargyris et al., 2023; Pati et al., 2022; Collaborative, 2021). Intended as orientation, not a formal ranking. Source: authors.

each task, calibration reliability plots, decision-curve analyses, per-landmark AUC(t) and lead-time plots for T1 and T2, and subgroup AUROC plots for T3–T5.

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